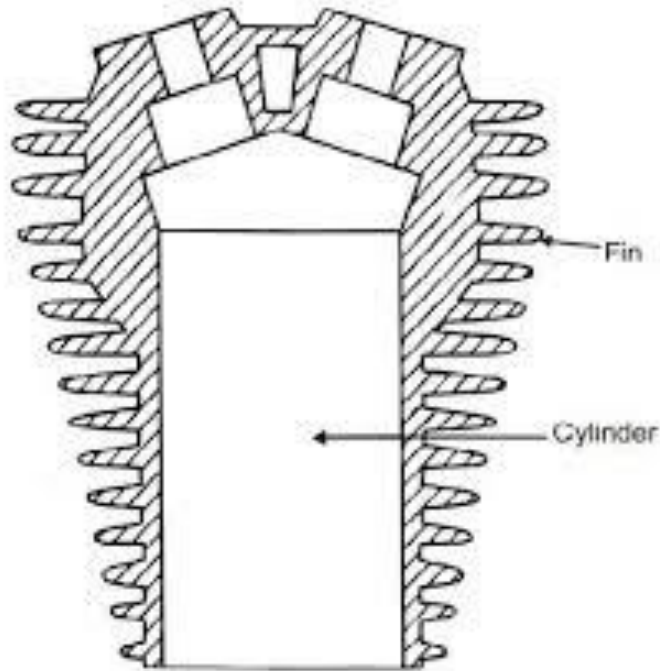




THERMAL ANALYSIS OF IC ENGINE CYLINDER BLOCK USING DIFFERENT MATERIALS AND COMPARISON OF RESULTS

□ CONTENT

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- Cooling system
- Methodology
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□ Objective

- The Engine cylinder is one of the major automobile components, which is subjected to high temperature variations and thermal stresses. In order to cool the cylinder, fins are provided on the surface of the cylinder to increase the rate of heat transfer. By doing thermal analysis on the engine cylinder fins, it is helpful to know the heat dissipation inside the cylinder.
- It is commonly used in gasoline engines, has the disadvantage of not being replaceable. When excessive wear occurs in a block of this type, the cylinder must be re-bored or honed. Reconditioning of this type cannot be repeated indefinitely and, in time, the entire block must be replaced. Another disadvantage is the inconvenience, especially in large engines, of having to remove the entire cylinder block from a ship in order to recondition the cylinders

Introduction

- The cylinder is one of the most important engine components that is subjected to extreme temperature changes and thermal stress
- Fins on the surface of the cylinder aid in cooling and heat transfer. Fins are mechanical structures that use convection to cool the engine.
- Extending the fins on an IC engine is widely accepted as a way to improve heat transfer.

Cooling System

➤ Types of Cooling System

- There are primarily Two types of cooling systems:
 1. Air cooled system
 2. Water cooled system.
- An automotive cooling system must perform several functions
 - Remove excess heat from the engine
 - Maintain a consist engine temperature
 - Help a cold engine warm-up quickly
 - Provide a means of warming the passenger compartment

□ Methodology

➤ Steps of working method

Step 1: Aggregation data and information associated with cooling fins of IC engines.

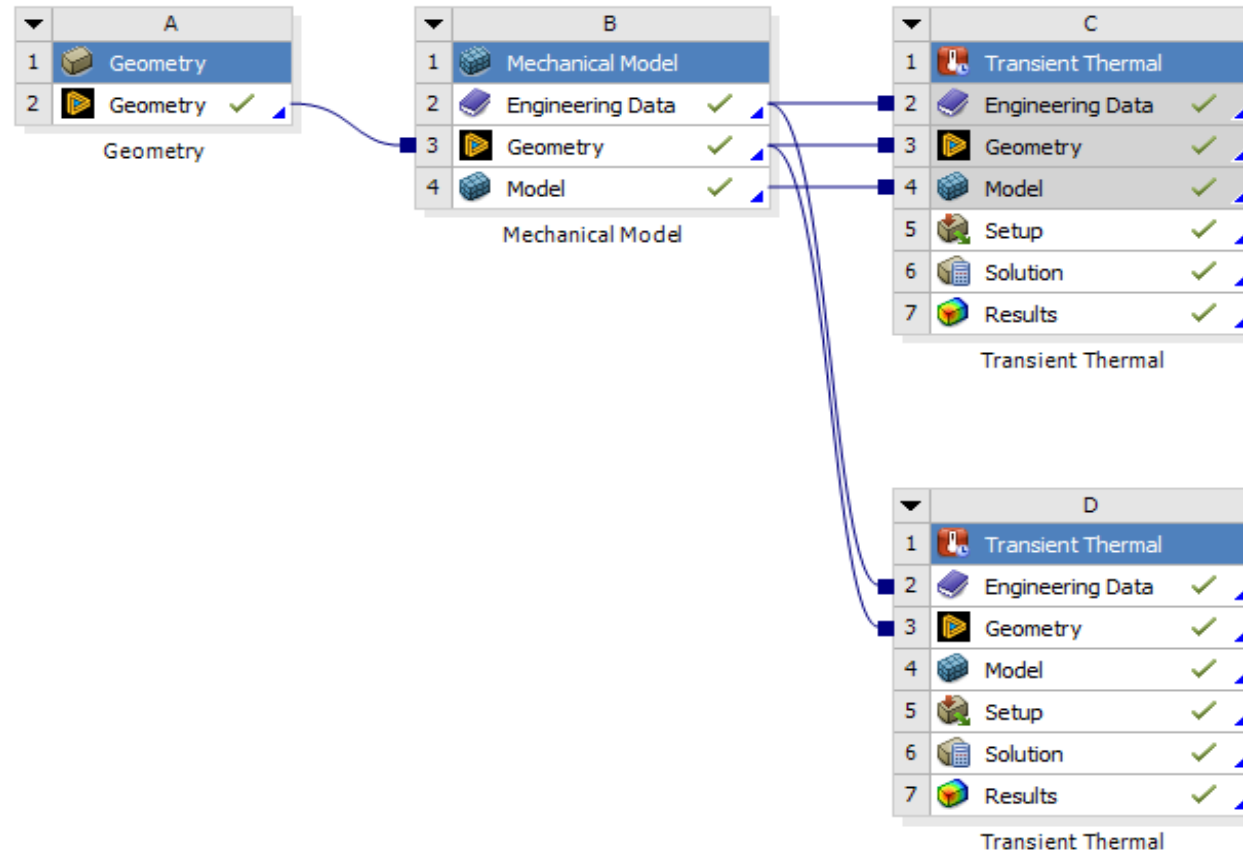
Step 2: A completely parametric model of the cylinder block with fin is formed in Solidworks software package.

Step 3: Model obtained in Step a pair of is analyzed using ANSYS 22.0 (Workbench), to get the warmth or heat rate , thermal gradient and nodal temperatures.

Step 4: Manual calculations are done.

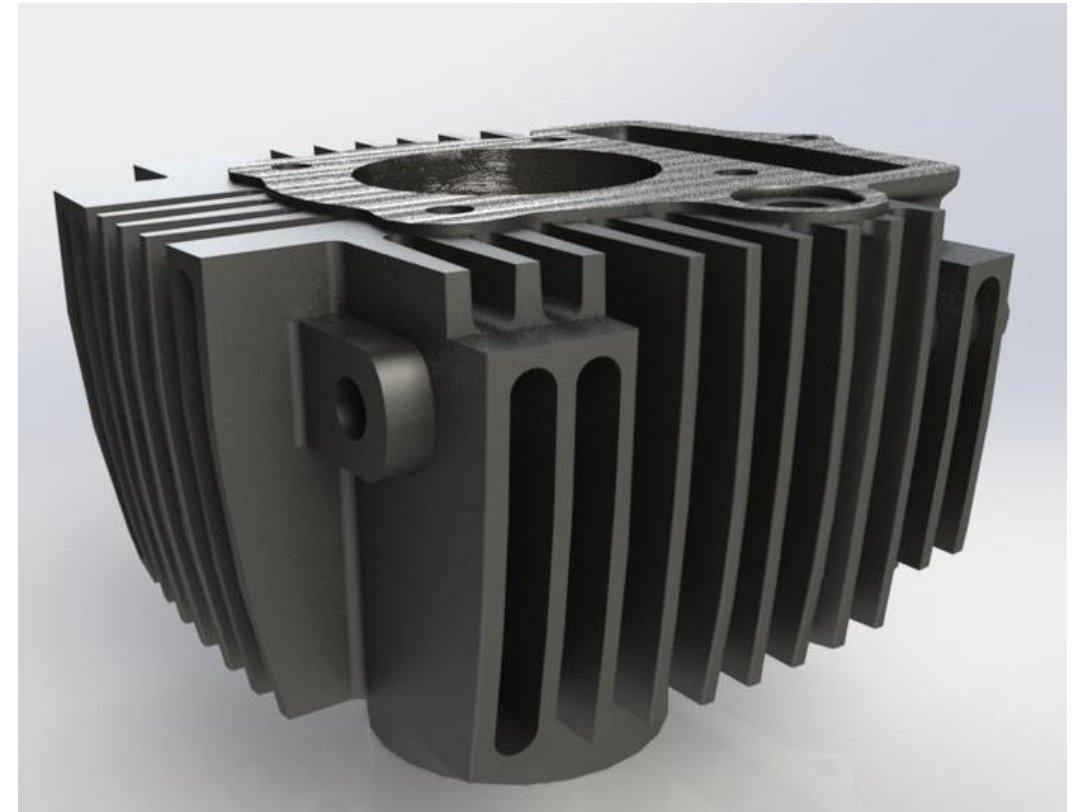
Step 5: Finally, we tend to compare the results obtained from ANSYS for completely different material.

□ Total Workflow

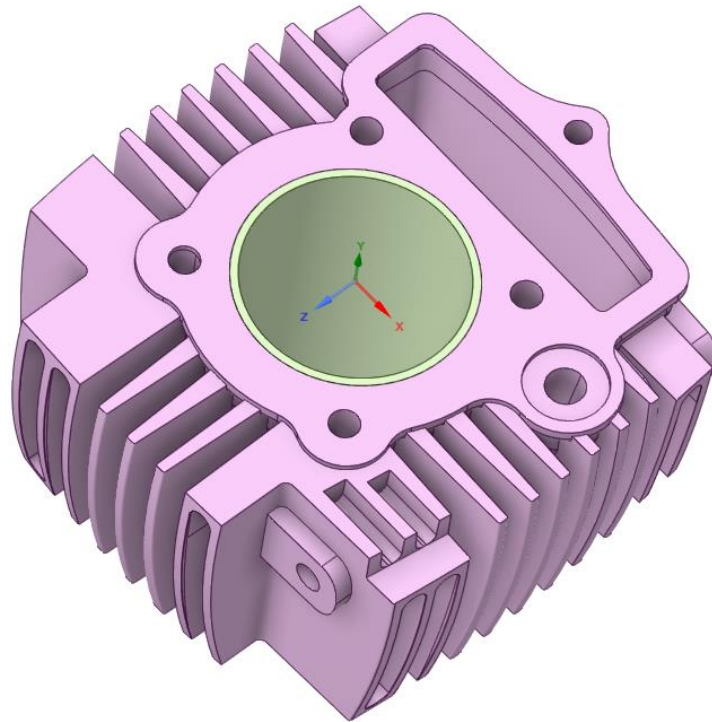


□ Engine Specifications

Specifications	Dimensions
Displacement	97.50 cubic cm (5.95 cubic inches)
Engine type	Single cylinder, four-stroke
Power	7.37 HP (5.4 kW)) @ 8000 RPM
Torque	7.95 Nm (0.8 kgf-m or 5.9 ft.lbs) @ 5000 RPM



□ CAD Model



□ Material Properties

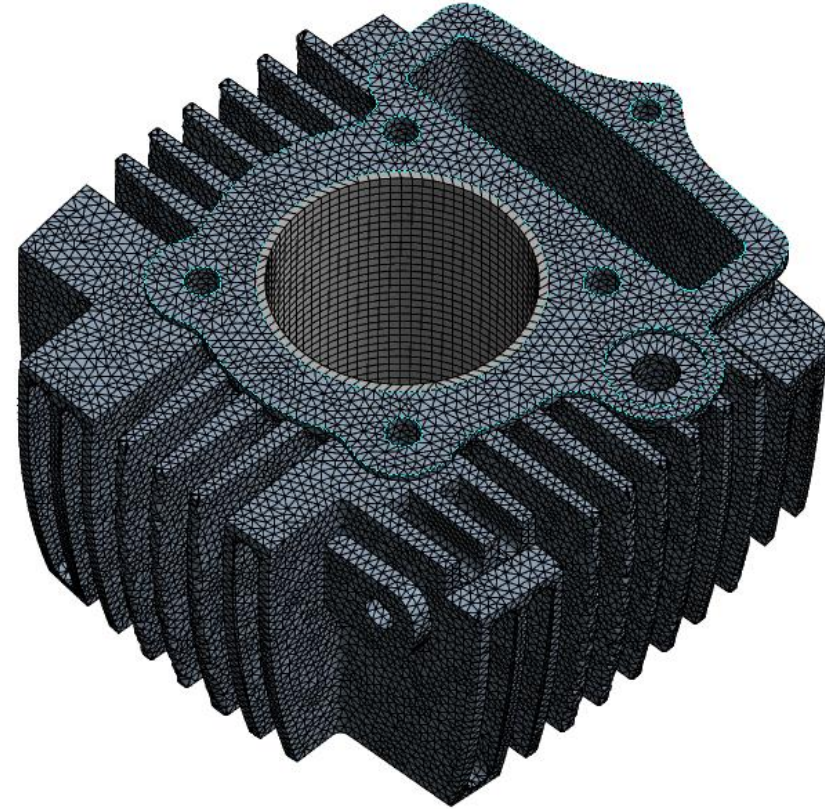
Sr.No.	Property	Material: Aluminium Alloy Gr. 6061	Gray Cast Iron	Titanium Alloy
1	Thermal Conductivity K	155.3 W/mk	52 W/mk	21.9 W/mk
2	Co efficient of Thermal expansion	2.78×10^{-5} K	1.1×10^{-5} K	9.4×10^{-6} K
3	Specific Heat	915 J/Kg K	447 J/Kg K	522 J/Kg K
4	Density	2.7 g/cc	7.2 g/cc	4.62 g/cc
5	Ambient temperature	295 K	295 K	295 K
6	Film Coefficient	$5 \times 10^6 \text{ W/mm}^2\text{°C}$	$5 \times 10^6 \text{ W/mm}^2\text{°C}$	$5 \times 10^6 \text{ W/mm}^2\text{°C}$
7	Poisson Ratio	0.33	0.28	0.36

□ Boundary condition

1. The maximum temperature generated inside the cylinder block is taken as 1000 °C.
2. For the meshing we used Programme control mechanical method using element size 2 mm.
3. Initial temperature 140 °C (Assumed).
4. Ambient Temperature 22 °C.
5. Convection Coefficient type is Average Film Temperature.
6. Heat Flow 100 W (Calculated by manual)

□ Meshing

- The mesh created in this work is shown in figure. Total node is generated 665435 & Total No. of Elements is 414332. Types of element used in the present work is Element Size .





Ansys Projects

Keywords

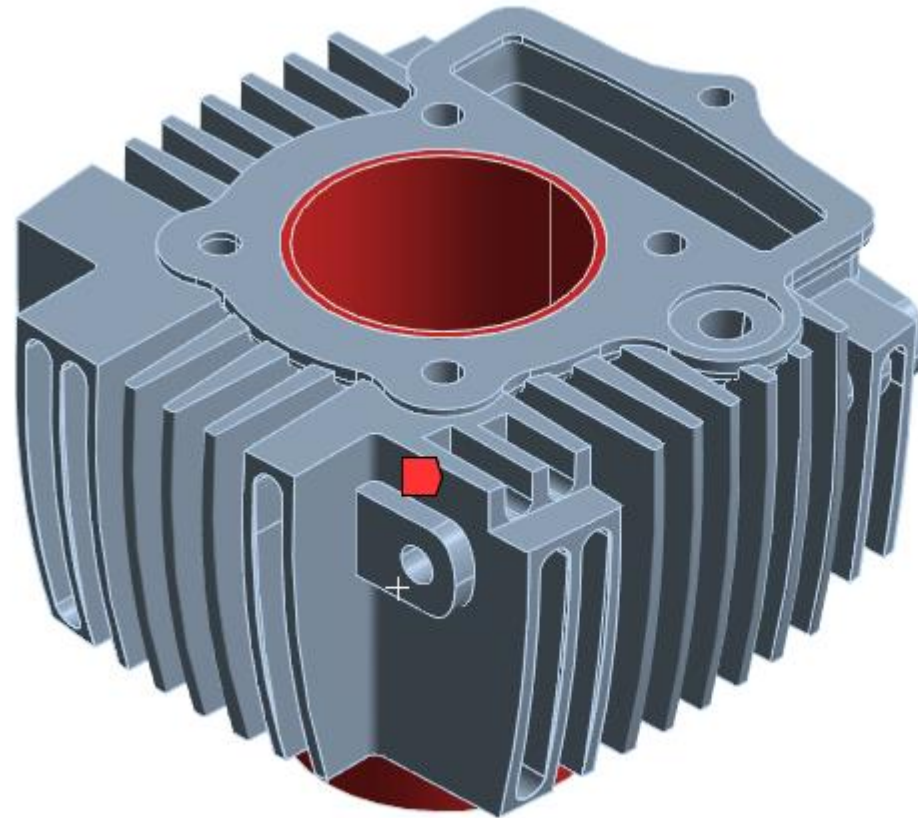
#ANSYS Workbench
#SpaceClaim
#Cut Geometry

“Engine cylinder block with fins”

ANSYS Mechanical
Meshing

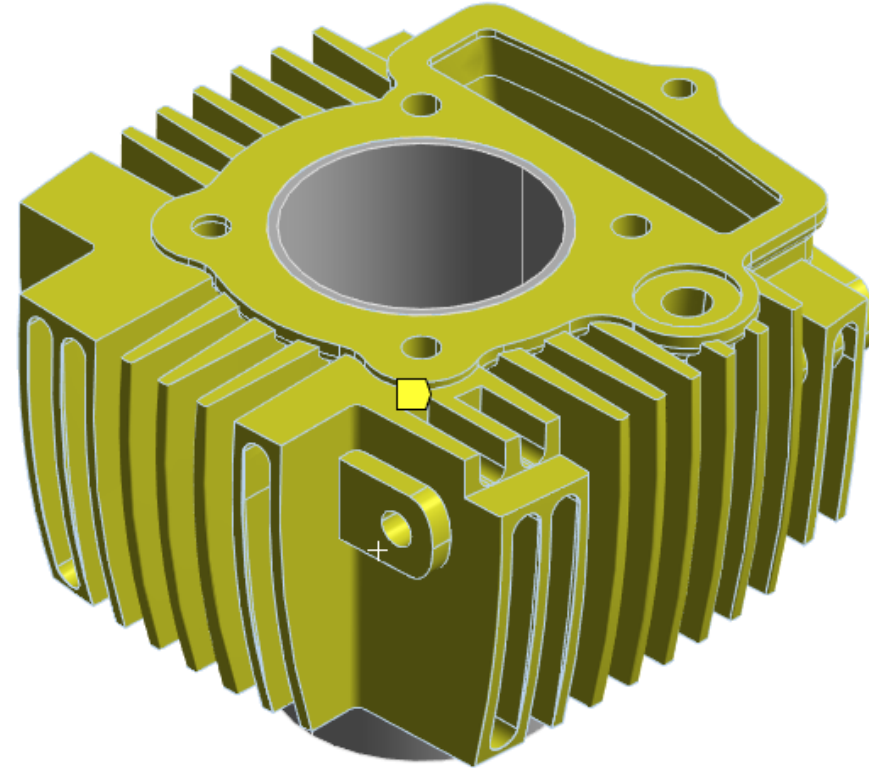
□ Boundary Condition

Temperature



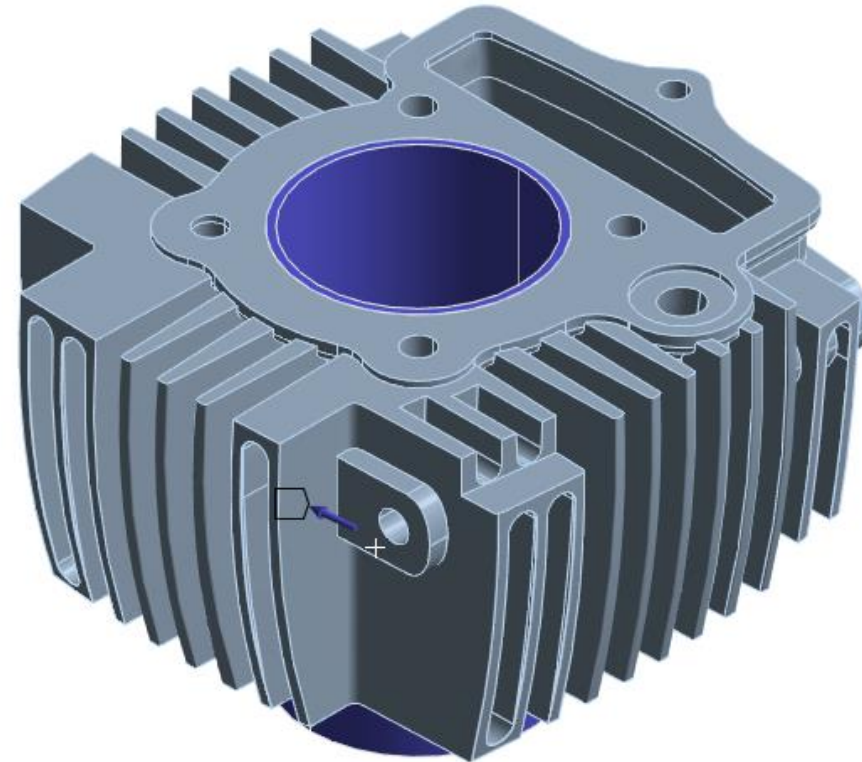
□ Boundary Condition

Convection



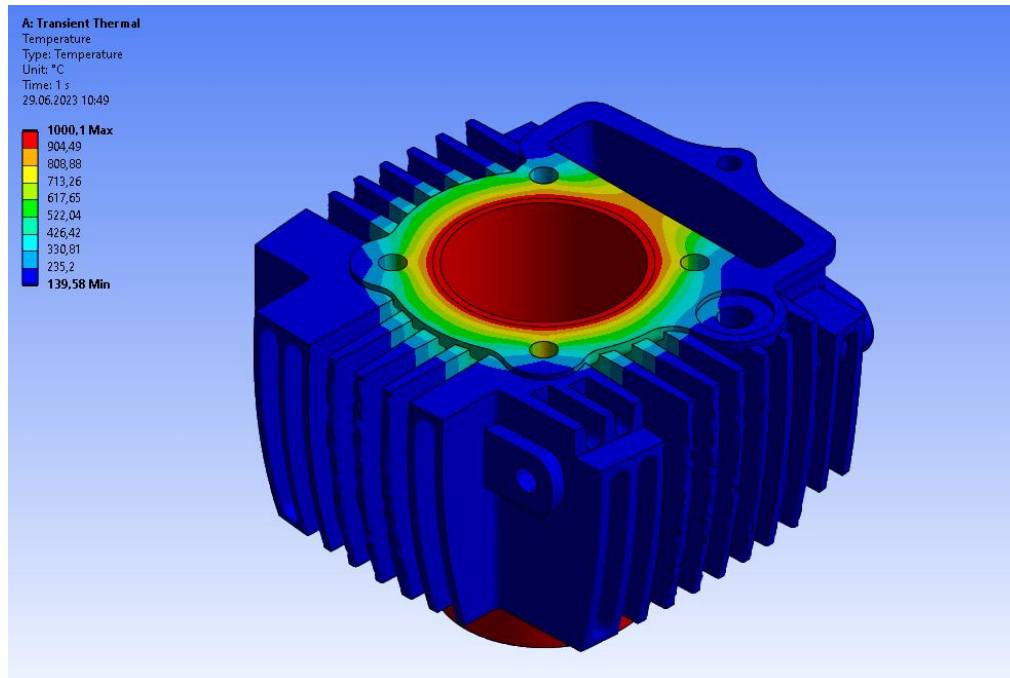
□ Boundary Condition

Heat Flow

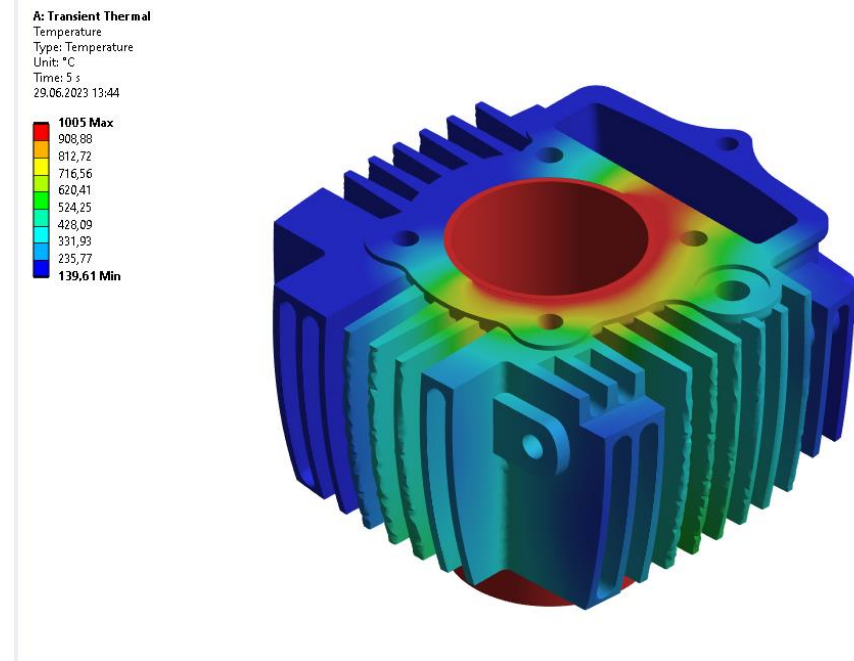


Comparison

Temperature Difference



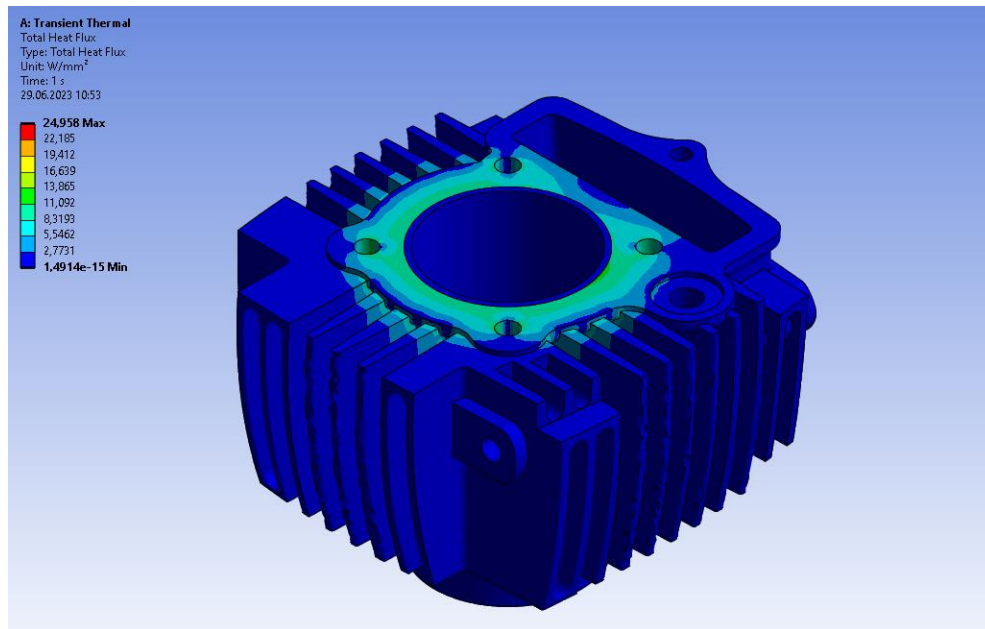
Cast Iron



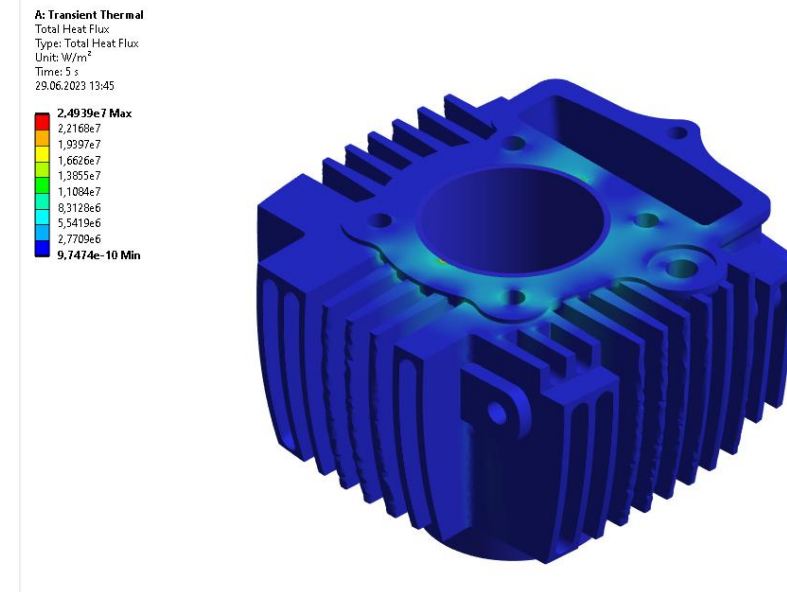
Titanium Alloy

Comparison

Total Heat Flux Difference



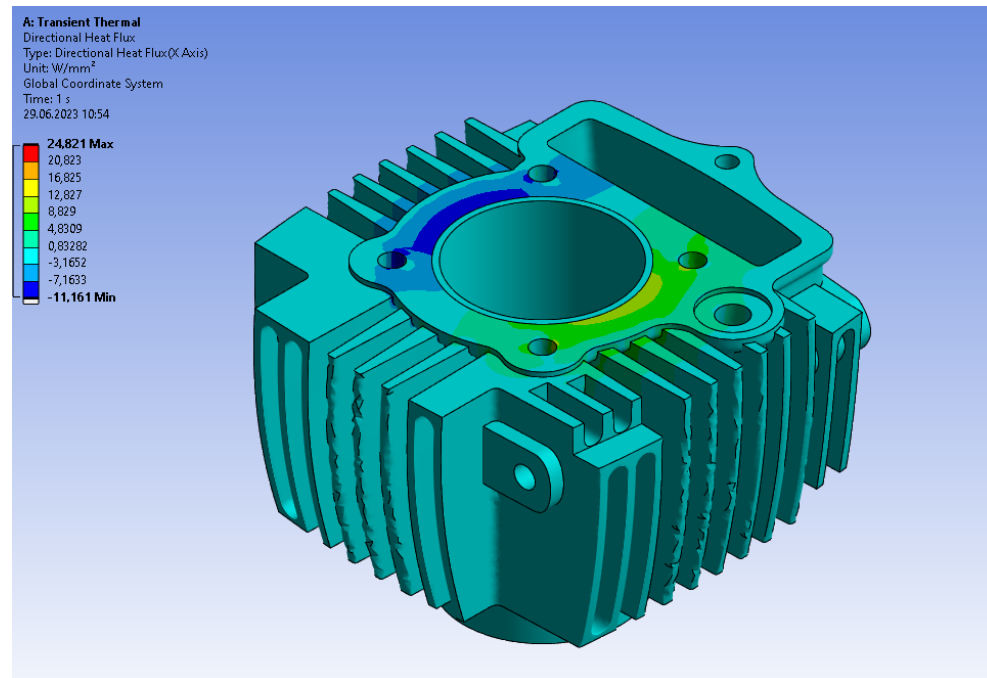
Cast Iron



Titanium Alloy

Comparison

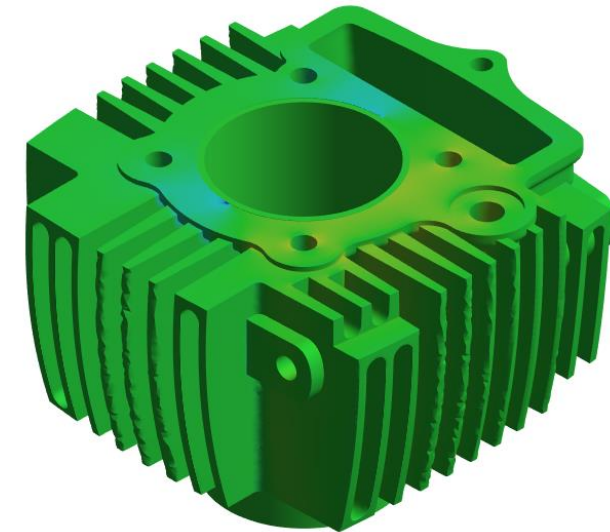
Directional Heat Flux Difference



Cast Iron

A: Transient Thermal
Directional Heat Flux
Type: Directional Heat Flux(X Axis)
Unit: W/m²
Global Coordinate System
Time: 5 s
29.06.2023 13:45

1,7811e7 Max
1,3945e7
1,0078e7
6,211e6
2,3442e6
-1,5227e6
-5,3895e6
-9,2563e6
-1,3123e7
-1,699e7 Min



Titanium Alloy

□ Conclusion

1. **Weight Reduction:** Titanium is significantly lighter than cast iron. This weight reduction can contribute to overall engine weight reduction, leading to improved fuel efficiency, lower emissions, and better vehicle performance.
2. **Corrosion Resistance:** Titanium exhibits exceptional corrosion resistance, especially in aggressive environments such as high-temperature and high-pressure conditions found within engine cylinders.
3. **Heat Transfer:** Titanium has excellent thermal conductivity properties, allowing it to dissipate heat effectively. This property is particularly beneficial for engine cylinders as it aids in efficient cooling, preventing overheating and potential engine damage.
4. **Thermal expansion compatibility:** Titanium has a coefficient of thermal expansion that is closer to that of aluminum, which is commonly used in engine blocks. This compatibility reduces the risk of thermal stress and potential issues associated with differential expansion between the cylinder liner and the engine block.
5. **Design flexibility:** Titanium's excellent formability and machinability allow for complex designs and tighter tolerances in cylinder liner manufacturing.

□ Reference

- Ansys ACP User's Guide:-
- Link:
https://ansyshelp.ansys.com/account/secured?returnurl=/Views/Secured/corp/v231/en/acp_ug/acp_ug.html
- International Research Journal of Engineering and Technology (IRJET)
- Link: www.irjet.net
- <https://www.youtube.com/watch?v=4Jj0s-DAvfg&t=575s>

THANK YOU